| Ex - 07 | Data Visualization Using Power BI |
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**AIM**

To visualize data using Power BI by connecting data sources, creating charts, applying DAX calculations, and building an interactive dashboard.

**PROBLEM STATEMENT**

To analyze data using Power BI by creating bar charts, line charts, pie charts, calculated columns, and measures to present meaningful insights.

**PROCEDURE**

1. Open Microsoft Power BI Desktop and explore the Report View, Data View, Model View, and Visualizations pane.

2. Connect to an Excel or CSV dataset using Home → Get Data, select the required file, and click Load to import the data.

3. Connect to a SQL database by selecting Home → Get Data → SQL Server, entering the server details, and selecting the required tables.

4. Open Power Query Editor using Transform Data to remove duplicates, handle missing values, rename columns, and change data types.

5. Create a bar chart by selecting the Clustered Bar Chart visual and dragging the category and numerical fields into the appropriate field wells.

6. Create a line chart by selecting the Line Chart visual, placing the date field on the X-axis and the sales or numerical field on the Y-axis to analyze trends.

7. Create a pie chart by selecting the Pie Chart visual, placing the category field in the Legend and the numerical field in Values to display proportions.

8. Create a calculated column using Table tools → New column and enter a DAX formula to calculate additional values for each row.

9. Create measures using Table tools → New measure and apply DAX functions such as SUM() and AVERAGE() to calculate totals and averages.

10. Build an interactive dashboard-style report by arranging charts, KPI cards, and slicers, formatting titles and colors, testing filters, and saving the report as a .pbix file.

**IMPLEMENTATION**

1. Calculated Column (DAX)

```
Total Amount = Sales[Sales] * Sales[Quantity]
```

2. Measure – Total Sales

```
Total Sales = SUM(Sales[Total Amount])
```

3. Measure – Average Sales

```
Average Sales = AVERAGE(Sales[Total Amount])
```

**Dashboard components:**

- Bar chart – Product-wise sales
- Line chart – Monthly sales trend
- Pie chart – Category-wise sales
- Card – Total sales
- Slicer – Filter by category

Note: Replace the table and column names with those in your dataset.




**RESULT**

Data was successfully visualized using Power BI. Bar charts, line charts, pie charts, DAX calculated columns, and measures were created and combined into an interactive dashboard for effective data analysis.